IA07: Test Evidence, PR Review and CI Walkthrough

Identity: jpkanquandah, team Intellectuals. Repositories: https://github.com/Dept-of-Comp-Sci-University-of-Ghana/DE-IND-jpkanquandah and https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-intellectuals. Issue: (link in the Sakai portfolio). PR: (link in the Sakai portfolio). Date: 23 October 2026.

Feature context

- **User:** customer on a phone. **Feature:** Enquire button opens WhatsApp with a prepared message.
- **Requirement and story:** NFR-12 (right number), story "As a customer, I want to tap Enquire…".
- **Acceptance criteria:** AC1 right number and service name; AC2 fallback when WhatsApp cannot open (see IA03).
- **Client value:** enquiries reach the owner without repeated DMs.
- **Status (honest):** the Enquire button is implemented on the deployed site; the automated link-builder tests below are planned checks unless marked otherwise.

Six test cases

	Type	Starting condition	Action	Test data	Expected result	Status and evidence
	Normal	Services section open	Tap Enquire on a service	Service "Mobile Crane Hire (sample)"	WhatsApp link has the approved number and the service name	Planned (not yet run by me)
	Invalid input	Link builder called with an unknown id	Call buildWhatsAppLink	id "xyz"	General message link; no crash	Planned (not yet run by me)
	Edge	Service name with special characters	Build the link	"Heavy & Wide Loads: 50% off"	Message is correctly URL-encoded, link still opens	Planned (not yet run by me) (TC-08)
	Authorization or security	A change edits the number	Open a PR changing the number	Fake number 233111111111	Review is required; tests fail if it is not the approved number (TC-19)	Planned (not yet run by me)
	Failure or dependency	Phone without WhatsApp	Tap Enquire	Any service	Phone shows WhatsApp page; Contact section still shows the number	Planned (not yet run by me)
	Regress	After a	Open the	None	Home opens and all	Planned

	Type	Starting condition	Action	Test data	Expected result	Status and evidence
	ion	navigation change	phone menu and tap Home		Enquire buttons still work	(not yet run by me)

Safe test data

All test data is synthetic: made-up service names, the number 233000000000 and a fake number for the security test. I chose special characters on purpose to test encoding. No real client records, real customer chats or credentials are used.

PR review and CI

Planned PR review checklist (to be used on the Enquire button pull request, (link in the Sakai portfolio), or on a future PR):

- ☐ Requirement fit: the change matches the story and acceptance criteria.
- ☐ Validation: ids, number and message are handled.
- ☐ Tests: new or changed code has tests and they pass.
- ☐ Data risk: no customer data or real private details added.
- ☐ Security: number unchanged unless approved; no new third-party scripts.
- ☐ Documentation: README or content notes updated.
- ☐ AI disclosure: the PR says whether AI was used.
- ☐ System impact: nothing else (menu, images) breaks.

CI: our team pipeline is meant to run lint, test and build on every pull request, and make test runs the same checks locally. Commands to run: make setup, make lint, make test. No CI run log is attached to this report and I am not claiming a successful run. The team must improve: add a test job for the link builder that fails if a link does not start with the approved number, and require CI to pass before merging.

Defect walkthrough

Likely defect: a service name with an ampersand or percent sign is not encoded, so the message is cut off in WhatsApp.

- **Reproduction:** put a service called “Heavy & Wide Loads” in the content, tap Enquire.
- **User-visible result:** the chat opens with only “Hello, I would like to ask about Heavy”.
- **Evidence:** the generated link in the browser address bar, the failing unit test.
- **Fix:** encode the message with encodeURIComponent and add test 3 so it cannot come back.

Video and AI disclosure

- Video (3 to 4 minutes, covering feature, tests, PR focus, CI and one weakness): https://youtu.be/bMEDxWQgDD4. Files in IA07/.

- **AI used:** Yes. Claude (Anthropic), 23 October 2026: drafted the test table and checklist. I changed the test data to synthetic values and marked every test as planned because I have not run them. My own understanding: the Enquire link is only as safe as the number and encoding behind it, so tests should check both.